

13.1 Bilinear Forms and Their Matrices

Chapter 6 needed a function of several vectors that was linear in each of them separately, and called it a multilinear form (Definition 6.2.1). Chapter 10 needed a function of two vectors that was linear in the first, conjugate-linear in the second, and positive on the diagonal, and called it an inner product. This section takes the common skeleton of the two — two slots, linear in each, nothing else assumed — and studies it on its own. The payoff is immediate: such a function is recorded by a square matrix, exactly as a linear map is, and the whole rest of the chapter is about how that matrix behaves.

Throughout, V is a vector space over a field F . From the second subsection on, V is finite-dimensional with $\dim V = n \geq 1$, and $\mathcal{B} = (\mathbf{v}_1, \dots, \mathbf{v}_n)$ is an ordered basis of V . **No hypothesis on the characteristic of F is used anywhere in this section**, and none is used in the next one either.

13.1.1 Pairing without measuring

The dot product on F^n is a machine with two inputs and one scalar output. Over $\mathbb{R}$ it does two different jobs. It **pairs**: it takes $\mathbf{x}$ and $\mathbf{y}$ and returns $\sum_i x_i y_i$, linearly in each of them. And it **measures**: $\mathbf{x} \cdot \mathbf{x}$ is a sum of squares, so it is positive unless $\mathbf{x} = \mathbf{0}$, and from that one fact Chapter 10 extracted lengths, angles, orthonormal bases and everything built on them.

Measuring is the fragile half. It needs an order on F , so it is meaningless over $\mathbb{C}$ without a conjugation and meaningless over $\mathbb{F}_5$ altogether. Pairing needs nothing: $\sum_i x_i y_i$ makes sense over every field, and so do plenty of other two-slot expressions, such as $x_1 y_2 - x_2 y_1$ or $x_1 y_1 - x_2 y_2$. Chapter 6 already met the first of these as the signed area, and met the general notion as the case $k = 2$ of a multilinear form. We give that case its own name, because from here on it is the object of study rather than a step toward the determinant.

A bilinear form pairs two vectors of the same space into a scalar, linearly in each slot separately.

Definition 13.1.1. Bilinear Form

Let V be a vector space over a field F . A **bilinear form** on V is a function $\beta: V \times V \rightarrow F$ that is linear in each argument when the other is held fixed: **for all** $\mathbf{u}, \mathbf{u}', \mathbf{v}, \mathbf{v}' \in V$ and **all** $c \in F$,

$$(B1) \quad \beta(\mathbf{u} + \mathbf{u}', \mathbf{v}) = \beta(\mathbf{u}, \mathbf{v}) + \beta(\mathbf{u}', \mathbf{v}) \text{ and } \beta(c\mathbf{u}, \mathbf{v}) = c\beta(\mathbf{u}, \mathbf{v});$$

$$(B2) \quad \beta(\mathbf{u}, \mathbf{v} + \mathbf{v}') = \beta(\mathbf{u}, \mathbf{v}) + \beta(\mathbf{u}, \mathbf{v}') \text{ and } \beta(\mathbf{u}, c\mathbf{v}) = c\beta(\mathbf{u}, \mathbf{v}).$$

Equivalently, β is a 2-linear form on V in the sense of Definition 6.2.1.

In words: (B1) says that for each fixed $\mathbf{v}$, the function $\mathbf{u} \mapsto \beta(\mathbf{u}, \mathbf{v})$ is a linear functional on V (Definition 4.1.1), and (B2) says the same for $\mathbf{v} \mapsto \beta(\mathbf{u}, \mathbf{v})$ with $\mathbf{u}$ fixed. The word **fixed** is the whole content. Nothing is assumed about how the two slots compare: $\beta(\mathbf{u}, \mathbf{v})$ and $\beta(\mathbf{v}, \mathbf{u})$ may be unrelated, and $\beta(\mathbf{v}, \mathbf{v})$ may be any scalar at all, including 0 for a non-zero $\mathbf{v}$.

Two consequences come free from linearity in one slot. Taking $c = 0$ in (B1) gives $\beta(\mathbf{0}, \mathbf{v}) = 0$ for every $\mathbf{v}$, and likewise $\beta(\mathbf{u}, \mathbf{0}) = 0$. And by induction on the number of terms,

$$\beta\left(\sum_i a_i \mathbf{u}_i, \sum_j b_j \mathbf{v}_j\right) = \sum_{i,j} a_i b_j \beta(\mathbf{u}_i, \mathbf{v}_j),$$

which is the identity everything in this section rests on: a bilinear form is determined by its

values on pairs drawn from a spanning set.

Example. Six two-slot functions

Decide which of the following are bilinear forms on the given space over the given field.

- (a) The dot product on F^n : $\beta(\mathbf{x}, \mathbf{y}) = x_1y_1 + \cdots + x_ny_n$.
- (b) On F^n , for a fixed $\mathbf{A} \in M_n(F)$: $\beta_{\mathbf{A}}(\mathbf{x}, \mathbf{y}) = \mathbf{x}^\top \mathbf{A} \mathbf{y}$.
- (c) On F^2 : $\beta(\mathbf{x}, \mathbf{y}) = x_1y_2 - x_2y_1$.
- (d) On $\mathbb{R}[x]_{\leq 2}$: $\beta(f, g) = \int_0^1 f(t)g(t) dt$.
- (e) On any V : the zero function $\beta(\mathbf{u}, \mathbf{v}) = 0$.
- (f) On $\mathbb{R}^n$: $\beta(\mathbf{x}, \mathbf{y}) = \|\mathbf{x}\| \|\mathbf{y}\|$, the product of the Euclidean lengths.

Solution.

- (a) Fix $\mathbf{y}$. Then $\mathbf{x} \mapsto \sum_i x_i y_i$ is a linear functional on F^n , since it is a fixed linear combination of the coordinate functionals (Theorem 4.1.1). Fixing $\mathbf{x}$ gives the same statement in $\mathbf{y}$. So (B1) and (B2) hold and the dot product is bilinear.
- (b) With $\mathbf{A} = \mathbf{I}_n$ this is (a). In general, write out the entries: $\mathbf{x}^\top \mathbf{A} \mathbf{y} = \sum_{i,j} a_{ij} x_i y_j$. For fixed $\mathbf{y}$ this is $\sum_i (\sum_j a_{ij} y_j) x_i$, a fixed linear combination of the coordinates of $\mathbf{x}$, hence a linear functional; symmetrically in $\mathbf{y}$. So $\beta_{\mathbf{A}}$ is bilinear for **every** $\mathbf{A}$, symmetric or not. Part (c) below is the case $\mathbf{A} = \begin{pmatrix} 0 & 1 \\ -1 & 0 \end{pmatrix}$.
- (c) This is the signed area of Chapter 6, and it is bilinear by Example (Alternating Forms) (a). It is worth noticing what it does on the diagonal: $\beta(\mathbf{x}, \mathbf{x}) = x_1x_2 - x_2x_1 = 0$ for **every** $\mathbf{x}$. A bilinear form may vanish identically on the diagonal without being the zero form.
- (d) For fixed g , the map $f \mapsto \int_0^1 fg$ is linear, because the integral of a sum is the sum of the integrals and constants come out. Symmetrically in g . So β is bilinear. Here $\beta(f, f) = \int_0^1 f^2 \geq 0$, so over $\mathbb{R}$ this form does measure: it is the inner product of Chapter 10.
- (e) Both clauses read $0 = 0 + 0$ and $0 = c \cdot 0$. The zero form is bilinear on every space over every field. It is the degenerate case, and it matters: it shows that a bilinear form can carry no information whatsoever, and it is the form whose matrix below is the zero matrix.
- (f) Not bilinear. Fix $\mathbf{y} = \mathbf{e}_1$. Then $\beta(\mathbf{e}_1, \mathbf{e}_1) + \beta(-\mathbf{e}_1, \mathbf{e}_1) = 1 + 1 = 2$, while $\beta(\mathbf{e}_1 + (-\mathbf{e}_1), \mathbf{e}_1) = \beta(\mathbf{0}, \mathbf{e}_1) = 0$. Additivity in the first slot fails, so (B1) fails. The homogeneity half of (B1) also fails, since $\beta(-\mathbf{e}_1, \mathbf{e}_1) = 1 \neq -\beta(\mathbf{e}_1, \mathbf{e}_1)$.

Item (f) is the non-example by minimal change worth remembering: the Cauchy–Schwarz inequality says that $|\mathbf{x} \cdot \mathbf{y}| \leq \|\mathbf{x}\| \|\mathbf{y}\|$, so $\|\mathbf{x}\| \|\mathbf{y}\|$ sits right next to the dot product numerically, and yet it is not a form at all, because the absolute value hidden in the norm destroys additivity. A second minimal change makes the same point more cheaply: $\beta(\mathbf{x}, \mathbf{y}) = x_1y_1 + 1$ fails (B1) because $\beta(\mathbf{0}, \mathbf{0}) = 1 \neq 0$.

Warning. Bilinear is not linear. A bilinear form is not a linear function of the pair $(\mathbf{u}, \mathbf{v})$, and the two properties are not even compatible in general. For the dot product on $\mathbb{R}^2$, doubling both arguments multiplies the value by 4: $\beta(2\mathbf{e}_1, 2\mathbf{e}_1) = 4$, not $2\beta(\mathbf{e}_1, \mathbf{e}_1) = 2$.

Sums may only be split **one slot at a time**, with the other slot held frozen. Chapter 6 gave this warning for multilinear forms in general; it is repeated here because it is the mistake readers make most often when a form is written with the familiar-looking symbol $\mathbf{x}^\top \mathbf{A} \mathbf{y}$.

Remark. The same definition with two different spaces, $\beta: V \times W \rightarrow F$ linear in each slot, is called a **bilinear pairing** of V and W . The basic example is the evaluation pairing $V^* \times V \rightarrow F$, $(\varphi, \mathbf{v}) \mapsto \varphi(\mathbf{v})$, which is linear in φ by the definition of the operations on V^* (Definition 4.1.2) and linear in $\mathbf{v}$ because φ is. We insist on $V = W$ in Definition 13.1.1 because the questions of this chapter — is $\beta(\mathbf{u}, \mathbf{v})$ equal to $\beta(\mathbf{v}, \mathbf{u})$, what is $\beta(\mathbf{v}, \mathbf{v})$, which vectors are orthogonal to which — only make sense when both slots hold vectors of the same space.

13.1.2 The matrix of a form

The displayed expansion above says that a bilinear form on a finite-dimensional space is determined by finitely many numbers: the values $\beta(\mathbf{v}_i, \mathbf{v}_j)$ on pairs of basis vectors. There are n^2 of them, they are indexed by a row and a column, and that is a matrix. Chapter 3 recorded a linear map by a matrix in exactly the same spirit (Definition 3.6.1), and the only difference is where the bases sit. There the two bases belonged to the input space and the output space. A form has no output space to speak of, but it has two **slots**, and each slot has to be fed a basis before the table can be written down. Throughout this chapter both slots hold vectors of the same V and we feed them the same basis, so the two choices coincide; the notation still records both, with the subscript naming the basis used on the first slot and the superscript the basis used on the second.

The matrix of a form is its table of values on pairs of basis vectors.

Definition 13.1.2. Matrix of a Bilinear Form

Let β be a bilinear form on a finite-dimensional V with ordered basis $\mathcal{B} = (\mathbf{v}_1, \dots, \mathbf{v}_n)$. The **matrix of β with respect to $\mathcal{B}$** , also called its **Gram matrix**, is the matrix $[\beta]_{\mathcal{B}}^{\mathcal{B}} \in M_n(F)$ with entries

$$([\beta]_{\mathcal{B}}^{\mathcal{B}})_{ij} := \beta(\mathbf{v}_i, \mathbf{v}_j).$$

In words: the entry in row i and column j is the value of β with the i -th basis vector in the **left** slot and the j -th in the **right** slot. The subscript is the basis read into the left slot and the superscript the basis read into the right one; here they are the same $\mathcal{B}$, and the notation writes it twice rather than leaving a slot empty. The order matters, because $\beta(\mathbf{v}_i, \mathbf{v}_j)$ and $\beta(\mathbf{v}_j, \mathbf{v}_i)$ need not agree; the convention chosen here is the one that makes the recovery formula below carry no transpose on the **matrix**.

Chapter 10 also called a matrix of pairings a Gram matrix, and the two conventions differ, so it is worth saying how. Definition 10.2.3 set $(\mathbf{G})_{ij} = \langle \mathbf{v}_j, \mathbf{v}_i \rangle$, with the indices crossed, because an inner product is conjugate-linear in its second slot and the crossing is what makes $\mathbf{G}$ Hermitian and $\mathbf{x}^* \mathbf{G} \mathbf{x}$ come out as a squared norm. A bilinear form has no conjugation to accommodate, so here the natural order is the uncrossed one, which is what makes the recovery formula below free of transposes. For a real symmetric form the two agree; in general each is the transpose of the other.

Nothing needs checking for well-definedness: once $\mathcal{B}$ is fixed, each entry is a single value of a function. What does need saying is that no information is lost.

Theorem 13.1.1. Forms and Matrices Correspond

Let V be an n -dimensional vector space over F with ordered basis $\mathcal{B}$, and let β be a bilinear form on V with $\mathbf{A} = [\beta]_{\mathcal{B}}^{\mathcal{B}}$.

- (a) For all $\mathbf{u}, \mathbf{v} \in V$,

$$\beta(\mathbf{u}, \mathbf{v}) = [\mathbf{u}]_{\mathcal{B}}^{\top} \mathbf{A} [\mathbf{v}]_{\mathcal{B}}.$$

- (b) $\mathbf{A}$ is the **only** matrix in $M_n(F)$ with this property.

- (c) The assignment $\beta \mapsto [\beta]_{\mathcal{B}}^{\mathcal{B}}$ is a bijection from the set of bilinear forms on V onto $M_n(F)$, and it respects sums and scalar multiples: $[\beta + \gamma]_{\mathcal{B}}^{\mathcal{B}} = [\beta]_{\mathcal{B}}^{\mathcal{B}} + [\gamma]_{\mathcal{B}}^{\mathcal{B}}$ and $[c\beta]_{\mathcal{B}}^{\mathcal{B}} = c[\beta]_{\mathcal{B}}^{\mathcal{B}}$.

Idea. Part (a) is the expansion $\beta(\sum a_i \mathbf{v}_i, \sum b_j \mathbf{v}_j) = \sum_{i,j} a_i b_j \beta(\mathbf{v}_i, \mathbf{v}_j)$ with the double sum recognized as a matrix product. For (b), feed the property a pair of basis vectors and watch it return a single entry. For (c), (b) gives injectivity, and surjectivity is [Example \(Six two-slot functions\)](#) (b) transported to V through coordinates.

Proof.

- (a) Write $[\mathbf{u}]_{\mathcal{B}} = (a_1, \dots, a_n)$ and $[\mathbf{v}]_{\mathcal{B}} = (b_1, \dots, b_n)$, so that $\mathbf{u} = \sum_i a_i \mathbf{v}_i$ and $\mathbf{v} = \sum_j b_j \mathbf{v}_j$. Expanding one slot at a time, by (B1) and then (B2),

$$\beta(\mathbf{u}, \mathbf{v}) = \sum_{i=1}^n \sum_{j=1}^n a_i b_j \beta(\mathbf{v}_i, \mathbf{v}_j) = \sum_{i,j} a_i a_{ij} b_j,$$

writing $a_{ij} = \beta(\mathbf{v}_i, \mathbf{v}_j)$ for the entries of $\mathbf{A}$. The last sum is the 1×1 matrix product $[\mathbf{u}]_{\mathcal{B}}^{\top} \mathbf{A} [\mathbf{v}]_{\mathcal{B}}$, by the definition of matrix multiplication (Definition 0.9.9).

- (b) Suppose $\mathbf{M} \in M_n(F)$ also satisfies $\beta(\mathbf{u}, \mathbf{v}) = [\mathbf{u}]_{\mathcal{B}}^{\top} \mathbf{M} [\mathbf{v}]_{\mathcal{B}}$ for all $\mathbf{u}, \mathbf{v}$. Take $\mathbf{u} = \mathbf{v}_i$ and $\mathbf{v} = \mathbf{v}_j$. Then $[\mathbf{v}_i]_{\mathcal{B}} = \mathbf{e}_i$ and $[\mathbf{v}_j]_{\mathcal{B}} = \mathbf{e}_j$, so the right-hand side is $\mathbf{e}_i^{\top} \mathbf{M} \mathbf{e}_j = m_{ij}$, while the left-hand side is $\beta(\mathbf{v}_i, \mathbf{v}_j) = a_{ij}$. Hence $m_{ij} = a_{ij}$ for all i, j , that is, $\mathbf{M} = \mathbf{A}$.

- (c) *Injective.* If $[\beta]_{\mathcal{B}}^{\mathcal{B}} = [\gamma]_{\mathcal{B}}^{\mathcal{B}}$, then by (a) the two forms agree at every pair $(\mathbf{u}, \mathbf{v})$, so $\beta = \gamma$.

Surjective. Given $\mathbf{M} \in M_n(F)$, define $\gamma(\mathbf{u}, \mathbf{v}) := [\mathbf{u}]_{\mathcal{B}}^{\top} \mathbf{M} [\mathbf{v}]_{\mathcal{B}}$. The coordinate map $\mathbf{v} \mapsto [\mathbf{v}]_{\mathcal{B}}$ is linear (Corollary 3.5.5), and $(\mathbf{x}, \mathbf{y}) \mapsto \mathbf{x}^{\top} \mathbf{M} \mathbf{y}$ is bilinear on F^n by [Example \(Six two-slot functions\)](#) (b); composing a bilinear form with a linear map in each slot leaves it bilinear, so γ is a bilinear form on V . By (b) its matrix is $\mathbf{M}$.

Linear. The forms $\beta + \gamma$ and $c\beta$ are defined pointwise, so their values at $(\mathbf{v}_i, \mathbf{v}_j)$ are $\beta(\mathbf{v}_i, \mathbf{v}_j) + \gamma(\mathbf{v}_i, \mathbf{v}_j)$ and $c\beta(\mathbf{v}_i, \mathbf{v}_j)$. These are the entries of $[\beta]_{\mathcal{B}}^{\mathcal{B}} + [\gamma]_{\mathcal{B}}^{\mathcal{B}}$ and of $c[\beta]_{\mathcal{B}}^{\mathcal{B}}$. This proves the theorem. ■

So bilinear forms on an n -dimensional space are the same thing as $n \times n$ matrices, once a basis is fixed — and in particular they form a vector space of dimension n^2 . The qualification “once a basis is fixed” is not decoration; the next section is entirely about what happens when the basis

is changed.

Example. Four Gram matrices

Compute $[\beta]_{\mathcal{B}}^{\mathcal{B}}$ in each case.

- (a) The dot product on F^n , with $\mathcal{B} = \mathcal{E}$ the standard basis.
- (b) $\beta_{\mathbf{A}}(\mathbf{x}, \mathbf{y}) = \mathbf{x}^{\top} \mathbf{A} \mathbf{y}$ on F^n , with $\mathcal{B} = \mathcal{E}$.
- (c) The signed area $\beta(\mathbf{x}, \mathbf{y}) = x_1 y_2 - x_2 y_1$ on F^2 , with $\mathcal{B} = \mathcal{E}$.
- (d) $\beta(f, g) = \int_0^1 fg$ on $\mathbb{R}[x]_{\leq 2}$, with $\mathcal{B} = (1, x, x^2)$.

Solution.

- (a) $\beta(\mathbf{e}_i, \mathbf{e}_j) = \delta_{ij}$, so $[\beta]_{\mathcal{E}}^{\mathcal{E}} = \mathbf{I}_n$. The dot product is the form whose matrix is the identity.
- (b) $\beta_{\mathbf{A}}(\mathbf{e}_i, \mathbf{e}_j) = \mathbf{e}_i^{\top} \mathbf{A} \mathbf{e}_j = a_{ij}$, so $[\beta_{\mathbf{A}}]_{\mathcal{E}}^{\mathcal{E}} = \mathbf{A}$. Together with [Theorem 13.1.1](#) (c) this says that on F^n with the standard basis, “form” and “matrix” are two words for the same data, and $\mathbf{x}^{\top} \mathbf{A} \mathbf{y}$ is the general form.
- (c) $\beta(\mathbf{e}_1, \mathbf{e}_1) = 0$, $\beta(\mathbf{e}_1, \mathbf{e}_2) = 1$, $\beta(\mathbf{e}_2, \mathbf{e}_1) = -1$, $\beta(\mathbf{e}_2, \mathbf{e}_2) = 0$, so

$$[\beta]_{\mathcal{E}}^{\mathcal{E}} = \begin{pmatrix} 0 & 1 \\ -1 & 0 \end{pmatrix}.$$

The matrix is not symmetric, and its diagonal is zero — the two features that will separate this form from the dot product throughout the chapter.

- (d) With $\mathcal{B} = (1, x, x^2)$ the (i, j) entry is $\int_0^1 t^{i-1} t^{j-1} dt = 1/(i+j-1)$, so

$$[\beta]_{\mathcal{B}}^{\mathcal{B}} = \begin{pmatrix} 1 & \frac{1}{2} & \frac{1}{3} \\ \frac{1}{2} & \frac{1}{3} & \frac{1}{4} \\ \frac{1}{3} & \frac{1}{4} & \frac{1}{5} \end{pmatrix}.$$

This matrix is symmetric, as it must be, since $\beta(f, g) = \beta(g, f)$. Its determinant is $1/2160$, so it is invertible; what that means for β is the subject of the last part of this section.

Quick check. A bilinear form β on F^2 satisfies $\beta(\mathbf{e}_1, \mathbf{e}_1) = 2$, $\beta(\mathbf{e}_1, \mathbf{e}_2) = -1$, $\beta(\mathbf{e}_2, \mathbf{e}_1) = 3$ and $\beta(\mathbf{e}_2, \mathbf{e}_2) = 0$. What is $\beta((1, 1), (1, -1))$?

Solution. The matrix is $\mathbf{A} = \begin{pmatrix} 2 & -1 \\ 3 & 0 \end{pmatrix}$, and by [Theorem 13.1.1](#) (a),

$$\beta((1, 1), (1, -1)) = \begin{pmatrix} 1 & 1 \end{pmatrix} \begin{pmatrix} 2 & -1 \\ 3 & 0 \end{pmatrix} \begin{pmatrix} 1 \\ -1 \end{pmatrix} = 6.$$

Expanding by hand gives the same. The pair is $(\mathbf{e}_1 + \mathbf{e}_2, \mathbf{e}_1 - \mathbf{e}_2)$, so the four terms are

$$\beta(\mathbf{e}_1, \mathbf{e}_1) - \beta(\mathbf{e}_1, \mathbf{e}_2) + \beta(\mathbf{e}_2, \mathbf{e}_1) - \beta(\mathbf{e}_2, \mathbf{e}_2),$$

which is $2 - (-1) + 3 - 0 = 6$. Note that the two off-diagonal values enter with **opposite** roles: $\beta(\mathbf{e}_1, \mathbf{e}_2)$ comes from the minus sign in the right slot, $\beta(\mathbf{e}_2, \mathbf{e}_1)$ from the plus sign in the left one. Since $\beta(\mathbf{e}_1, \mathbf{e}_2) \neq \beta(\mathbf{e}_2, \mathbf{e}_1)$ here, swapping them would change the answer.

Now the warning that the rest of the chapter exists to manage.

Warning. **The matrix of a form and the matrix of a map are different objects with the same name.** Both are square arrays attached to V and a basis, both are written $\mathbf{A}$, and both change when the basis changes — by **different rules**. If $\mathbf{P}$ is the change-of-coordinates matrix between the old and the new basis, then

$$\text{a map: } \mathbf{A} \mapsto \mathbf{P}^{-1}\mathbf{A}\mathbf{P}, \quad \text{a form: } \mathbf{A} \mapsto \mathbf{P}^\top\mathbf{A}\mathbf{P}.$$

The first is Theorem 3.7.3; the second is proved in the next section. They agree only when $\mathbf{P}^\top = \mathbf{P}^{-1}$. Concretely, take $\mathbf{A} = \text{diag}(1, 2)$ and $\mathbf{P} = \begin{pmatrix} 1 & 1 \\ 0 & 1 \end{pmatrix}$. Then

$$\mathbf{P}^{-1}\mathbf{A}\mathbf{P} = \begin{pmatrix} 1 & -1 \\ 0 & 2 \end{pmatrix}, \quad \mathbf{P}^\top\mathbf{A}\mathbf{P} = \begin{pmatrix} 1 & 1 \\ 1 & 3 \end{pmatrix}.$$

The first still has eigenvalues 1 and 2; the second has eigenvalues $2 \pm \sqrt{2}$. Same $\mathbf{A}$, same $\mathbf{P}$, two answers with nothing in common. Before writing $\mathbf{A}$ down, ask what it eats: a map eats one vector and returns a vector, so $\mathbf{A}\mathbf{x}$ is meaningful; a form eats two vectors and returns a scalar, so $\mathbf{x}^\top\mathbf{A}\mathbf{y}$ is meaningful and $\mathbf{A}\mathbf{x}$ is a meaningless intermediate. Almost every error in this chapter is this confusion.

13.1.3 The form as a map into the dual space

Fixing the right slot of β leaves a function of the left slot, and clause (B1) says that function is a linear functional. So each $\mathbf{v} \in V$ produces an element of V^* (Definition 4.1.2), and clause (B2) says the production is itself linear. That is a linear map $V \rightarrow V^*$, and it is the cleanest way to say what it means for a form to be non-degenerate.

Write $R_\beta: V \rightarrow V^*$ for the map sending $\mathbf{v}$ to the functional $\beta(\cdot, \mathbf{v})$, that is,

$$(R_\beta(\mathbf{v}))(\mathbf{u}) = \beta(\mathbf{u}, \mathbf{v}) \quad (\mathbf{u}, \mathbf{v} \in V).$$

The letter is local to this section and the subscript names the form; R is for the **right** slot, the one that is fixed. That $R_\beta(\mathbf{v})$ lies in V^* is (B1), and that R_β is linear is (B2). Its kernel has a name.

Definition 13.1.3. Radical of a Form

Let β be a bilinear form on V . The **radical** of β is

$$\text{rad}(\beta) := \{ \mathbf{v} \in V : \beta(\mathbf{u}, \mathbf{v}) = 0 \text{ for every } \mathbf{u} \in V \} = \ker R_\beta,$$

a subspace of V . Its elements are the vectors that **every** vector is orthogonal to.

Definition 13.1.4. Non-degenerate Form

A bilinear form β on V is **non-degenerate** if $\text{rad}(\beta) = \{\mathbf{0}\}$, that is, if for every **non-zero** $\mathbf{v} \in V$ there **exists** $\mathbf{u} \in V$ with $\beta(\mathbf{u}, \mathbf{v}) \neq 0$. Otherwise β is **degenerate**.

In words: non-degeneracy says that no non-zero vector is invisible to the form. It does not say that $\beta(\mathbf{v}, \mathbf{v}) \neq 0$, and it does not say that β is positive in any sense; it only says that a vector which the form cannot distinguish from $\mathbf{0}$ must be $\mathbf{0}$. On a finite-dimensional space it is a rank condition.

Proposition 13.1.2. Non-degeneracy is invertibility of the Gram matrix

Let β be a bilinear form on an n -dimensional V with ordered basis $\mathcal{B}$, and put $\mathbf{A} = [\beta]_{\mathcal{B}}^{\mathcal{B}}$. Then:

- (a) $\mathbf{A}$ is the matrix of R_{β} from the basis $\mathcal{B}$ to the dual basis $\mathcal{B}^*$; that is, $\mathbf{A} = [R_{\beta}]_{\mathcal{B}}^{\mathcal{B}^*}$.
- (b) $\dim \text{rad}(\beta) = n - \text{rank } \mathbf{A}$.
- (c) The following are equivalent: β is non-degenerate; $\mathbf{A}$ is invertible; R_{β} is an isomorphism $V \rightarrow V^*$.

Idea. Only (a) has content, and it is one line once the right formula is recalled: the coordinates of a functional in the dual basis are its values on $\mathcal{B}$ (Theorem 4.1.3 (c)), and the values of $R_{\beta}(\mathbf{v}_j)$ on $\mathcal{B}$ are the entries of the j -th column of $\mathbf{A}$. Parts (b) and (c) are then Rank–Nullity and the invertible matrix theorem, applied to a map we now know the matrix of.

Proof.

- (a) By Definition 3.6.1, the j -th column of $[R_{\beta}]_{\mathcal{B}}^{\mathcal{B}^*}$ is $[R_{\beta}(\mathbf{v}_j)]_{\mathcal{B}^*}$. By Theorem 4.1.3 (c), a functional $\varphi \in V^*$ has $\varphi = \sum_i \varphi(\mathbf{v}_i) \varphi_i$ in the dual basis $\mathcal{B}^* = (\varphi_1, \dots, \varphi_n)$, so the i -th coordinate of $R_{\beta}(\mathbf{v}_j)$ is $(R_{\beta}(\mathbf{v}_j))(\varphi_i) = \beta(\mathbf{v}_i, \mathbf{v}_j) = a_{ij}$. Hence the (i, j) entry of $[R_{\beta}]_{\mathcal{B}}^{\mathcal{B}^*}$ is a_{ij} , as claimed.
- (b) By (a) and Theorem 3.6.5, $\text{rank } R_{\beta} = \text{rank } \mathbf{A}$. Since $\text{rad}(\beta) = \ker R_{\beta}$, Rank–Nullity (Theorem 3.3.1) gives $\dim \text{rad}(\beta) = n - \text{rank } \mathbf{A}$.
- (c) By (b), $\text{rad}(\beta) = \{\mathbf{0}\}$ if and only if $\text{rank } \mathbf{A} = n$, which by Theorem 2.4.4 is exactly invertibility of $\mathbf{A}$. Since $\dim V^* = n = \dim V$ (Corollary 4.1.4), the map R_{β} is an isomorphism if and only if it is injective (Corollary 3.3.2 (e)), that is, if and only if its kernel is $\{\mathbf{0}\}$ (Theorem 3.2.3). This proves the proposition. ■

Remark. Freezing the **left** slot instead gives a second map $\mathbf{u} \mapsto \beta(\mathbf{u}, \cdot)$, whose kernel is the **left radical** $\{\mathbf{u} : \beta(\mathbf{u}, \mathbf{v}) = 0 \text{ for all } \mathbf{v}\}$. The same computation shows its matrix is $\mathbf{A}^{\top}$, so by Theorem 2.5.5 and Theorem 3.3.1 the two radicals have the same dimension — although, as [Exercise \(B3: Two radicals\)](#) shows, they need not be the same subspace. In particular “non-degenerate” means the same thing read from either side, and for the symmetric forms that occupy most of this chapter the two radicals coincide.

Example. Three radicals

Find $\text{rad}(\beta)$ and decide degeneracy.

- (a) $\beta(\mathbf{x}, \mathbf{y}) = x_1 y_1$ on F^2 .
- (b) $\beta(\mathbf{x}, \mathbf{y}) = x_1 y_1 - x_2 y_2$ on $\mathbb{R}^2$.
- (c) The zero form on any $V \neq \{\mathbf{0}\}$.

Solution.

- (a) $[\beta]_{\mathcal{E}}^{\mathcal{E}} = \begin{pmatrix} 1 & 0 \\ 0 & 0 \end{pmatrix}$, of rank 1, so the radical is one-dimensional by [Proposition 13.1.2](#) (b). Explicitly, $\beta(\mathbf{u}, \mathbf{v}) = u_1 v_1$ vanishes for all $\mathbf{u}$ exactly when $v_1 = 0$, so $\text{rad}(\beta) = \text{span}(\mathbf{e}_2)$. Degenerate.
- (b) $[\beta]_{\mathcal{E}}^{\mathcal{E}} = \text{diag}(1, -1)$, invertible, so β is non-degenerate. Yet $\beta((1, 1), (1, 1)) = 1 - 1 = 0$: a non-zero vector on which the form vanishes when paired **with itself**. Non-degeneracy forbids a vector orthogonal to everything; it says nothing about a vector orthogonal to itself. This form is the one this chapter returns to twice, in the sections on isotropic vectors and on Minkowski space.
- (c) The matrix is 0, of rank 0, and $\text{rad}(\beta) = V$. This is the extreme degenerate case: every vector is invisible.

Warning. **Non-degenerate does not mean definite, and the radical is not the set where $\beta(\mathbf{v}, \mathbf{v}) = 0$.** Example (b) above is non-degenerate, and the set $\{\mathbf{v} : \beta(\mathbf{v}, \mathbf{v}) = 0\}$ is the pair of lines $x_2 = \pm x_1$ — not a subspace, and not the radical, which is $\{\mathbf{0}\}$. Example (c) of [Example \(Six two-slot functions\)](#) has $\beta(\mathbf{v}, \mathbf{v}) = 0$ for **every** $\mathbf{v}$ and is still non-degenerate, since its matrix $\begin{pmatrix} 0 & 1 \\ -1 & 0 \end{pmatrix}$ is invertible. An inner-product-trained reader expects “ $\beta(\mathbf{v}, \mathbf{v}) = 0 \Rightarrow \mathbf{v} = \mathbf{0}$ ”, and that expectation is a theorem about positive definite forms over $\mathbb{R}$, not about forms.

Quick check. Let β be a bilinear form on a finite-dimensional V , and let $U = \text{rad}(\beta)$. Is β non-degenerate on the subspace U ?

Solution. No, unless $U = \{\mathbf{0}\}$ — and then only for a silly reason. The restriction of β to $U \times U$ is the **zero** form, since $\beta(\mathbf{u}, \mathbf{v}) = 0$ already for $\mathbf{u} \in V$ and $\mathbf{v} \in U$. The zero form on a non-zero space is degenerate, so the restriction is non-degenerate exactly when $U = \{\mathbf{0}\}$, that is, exactly when β was non-degenerate to begin with. The point of the question is that “non-degenerate” is a property of a form **on a specified space**, and restricting changes the space.

13.1.4 Exercises

13.1.4.1 A. Check your understanding

Exercise. A1

- (a) Define what it means for $\beta: V \times V \rightarrow F$ to be a bilinear form, stating both clauses in full.
- (b) Define the matrix $[\beta]_{\mathcal{B}}^{\mathcal{B}}$ of a bilinear form with respect to an ordered basis $\mathcal{B}$, and state how $\beta(\mathbf{u}, \mathbf{v})$ is recovered from it.
- (c) Determine whether the following is correct, with a reason: every bilinear form on F^n is $\beta(\mathbf{x}, \mathbf{y}) = \mathbf{x}^{\top} \mathbf{A} \mathbf{y}$ for exactly one $\mathbf{A} \in M_n(F)$.
- (d) Define the radical of β and state what it means for β to be non-degenerate.
- (e) Determine whether the following is correct, with a reason: if $\beta(\mathbf{v}, \mathbf{v}) = 0$ for some $\mathbf{v} \neq \mathbf{0}$, then β is degenerate.

Solution.

- (a) [Definition 13.1.1](#): linearity in the left slot for each fixed right argument, (B1), and linearity in the right slot for each fixed left argument, (B2).
- (b) [Definition 13.1.2](#): $([\beta]_{\mathcal{B}}^{\mathcal{B}})_{ij} = \beta(\mathbf{v}_i, \mathbf{v}_j)$. Recovery is [Theorem 13.1.1](#) (a): $\beta(\mathbf{u}, \mathbf{v}) = [\mathbf{u}]_{\mathcal{B}}^{\top} [\beta]_{\mathcal{B}}^{\mathcal{B}} [\mathbf{v}]_{\mathcal{B}}$.
- (c) Correct. Existence and uniqueness are [Theorem 13.1.1](#) (c) and (b) with $\mathcal{B} = \mathcal{E}$, for which $[\mathbf{x}]_{\mathcal{E}} = \mathbf{x}$.
- (d) [Definition 13.1.3](#) and [Definition 13.1.4](#): $\text{rad}(\beta) = \{\mathbf{v} : \beta(\mathbf{u}, \mathbf{v}) = 0 \text{ for all } \mathbf{u}\}$, and β is non-degenerate when this is $\{\mathbf{0}\}$.
- (e) Incorrect. The two conditions are unrelated. The signed area on F^2 has $\beta(\mathbf{v}, \mathbf{v}) = 0$ for every $\mathbf{v}$ and is non-degenerate, since its matrix $\begin{pmatrix} 0 & 1 \\ -1 & 0 \end{pmatrix}$ is invertible ([Proposition 13.1.2](#)).

13.1.4.2 B. Practice**Exercise. B1: Which are forms?**

Determine which of the following are bilinear forms on the given space. Justify your answer; for those that fail, name the clause that fails and give a witness.

- (a) On $\mathbb{R}^2$: $\beta(\mathbf{x}, \mathbf{y}) = x_1 y_1 + 2x_1 y_2 - 3x_2 y_1$.
- (b) On $\mathbb{R}^2$: $\beta(\mathbf{x}, \mathbf{y}) = x_1 y_1 + 1$.
- (c) On $\mathbb{R}[x]_{\leq 2}$: $\beta(f, g) = f(1)g(2) - f(0)g(0)$.
- (d) On $\mathbb{R}^2$: $\beta(\mathbf{x}, \mathbf{y}) = x_1^2 y_1$.
- (e) On $M_2(\mathbb{R})$: $\beta(\mathbf{X}, \mathbf{Y}) = \text{tr}(\mathbf{X}^{\top} \mathbf{Y})$.

Solution.

- (a) Bilinear: it is $\mathbf{x}^{\top} \mathbf{A} \mathbf{y}$ with $\mathbf{A} = \begin{pmatrix} 1 & 2 \\ -3 & 0 \end{pmatrix}$, so [Example \(Six two-slot functions\)](#) (b) applies.
- (b) Not bilinear. Clause (B1) fails: $\beta(\mathbf{0}, \mathbf{0}) = 1$, whereas homogeneity with $c = 0$ forces $\beta(\mathbf{0}, \mathbf{y}) = 0$ for any bilinear form.
- (c) Bilinear. Evaluation at a point is linear ([Theorem 0.8.5](#)), so for fixed g the map $f \mapsto g(2)f(1) - g(0)f(0)$ is a linear combination of two linear functionals, hence linear; symmetrically in g .

- (d) Not bilinear. Clause (B1) fails in the additive half: with $\mathbf{y} = \mathbf{e}_1$, $\beta(\mathbf{e}_1 + \mathbf{e}_1, \mathbf{e}_1) = 4$ while $\beta(\mathbf{e}_1, \mathbf{e}_1) + \beta(\mathbf{e}_1, \mathbf{e}_1) = 2$. The homogeneity half fails too, since $\beta(c\mathbf{x}, \mathbf{y}) = c^2 x_1^2 y_1$.
- (e) Bilinear. By Theorem 0.9.5 and linearity of the trace (Theorem 0.9.6), for fixed $\mathbf{Y}$ the map $\mathbf{X} \mapsto \text{tr}(\mathbf{X}^\top \mathbf{Y})$ is linear, and for fixed $\mathbf{X}$ the map $\mathbf{Y} \mapsto \text{tr}(\mathbf{X}^\top \mathbf{Y})$ is linear. This is the Frobenius inner product of Chapter 10, seen as a form.

Exercise. B2: A Gram matrix from sample values

On $V = \mathbb{R}[x]_{\leq 2}$, define $\beta(f, g) = f(0)g(0) + f(1)g(1) + f(2)g(2)$.

- (a) Verify that β is a bilinear form and compute $[\beta]_{\mathcal{B}}^{\mathcal{B}}$ for $\mathcal{B} = (1, x, x^2)$.
- (b) Compute $\beta(1 + x, x^2 - 1)$ directly from the definition, and again from the matrix. Check that the answers agree.
- (c) Hence decide whether β is non-degenerate.

Solution.

- (a) Each of the three terms is a product of two evaluations, hence bilinear as in [Exercise \(B1: Which are forms?\)](#) (c), and a sum of bilinear forms is bilinear by [Theorem 13.1.1](#) (c). Numbering rows and columns 1, 2, 3 for $1, x, x^2$, the (i, j) entry is $\sum_{t \in \{0, 1, 2\}} t^{i-1} t^{j-1}$, with $0^0 = 1$. So

$$[\beta]_{\mathcal{B}}^{\mathcal{B}} = \begin{pmatrix} 3 & 3 & 5 \\ 3 & 5 & 9 \\ 5 & 9 & 17 \end{pmatrix}.$$

- (b) Directly: $f = 1 + x$ takes the values 1, 2, 3 at 0, 1, 2, and $g = x^2 - 1$ takes the values $-1, 0, 3$. So $\beta(f, g) = 1(-1) + 2 \cdot 0 + 3 \cdot 3 = 8$. From the matrix, $[f]_{\mathcal{B}} = (1, 1, 0)$ and $[g]_{\mathcal{B}} = (-1, 0, 1)$, and

$$[\beta]_{\mathcal{B}}^{\mathcal{B}}[g]_{\mathcal{B}} = (2, 6, 12), \quad (1, 1, 0) \cdot (2, 6, 12) = 8.$$

- (c) Let $\mathbf{M}$ be the matrix with rows $(1, t, t^2)$ for $t = 0, 1, 2$. Then $[\beta]_{\mathcal{B}}^{\mathcal{B}} = \mathbf{M}^\top \mathbf{M}$, and $\det \mathbf{M} = 2$ by the Vandermonde formula (Theorem 6.8.1), so $\det[\beta]_{\mathcal{B}}^{\mathcal{B}} = 4 \neq 0$ by Theorem 6.5.4 and Theorem 6.4.5. By [Proposition 13.1.2](#), β is non-degenerate.

Exercise. B3: Two radicals

On F^3 , let $\beta(\mathbf{x}, \mathbf{y}) = \mathbf{x}^\top \mathbf{A} \mathbf{y}$ with

$$\mathbf{A} = \begin{pmatrix} 1 & 2 & 3 \\ 2 & 4 & 6 \\ 1 & 2 & 3 \end{pmatrix}.$$

Find $\text{rad}(\beta)$ and the left radical, give a basis of each, and decide whether they are equal.

Solution. Every row of $\mathbf{A}$ is a multiple of $(1, 2, 3)$, so $\text{rank } \mathbf{A} = 1$ and both radicals have dimension $3 - 1 = 2$, by Proposition 13.1.2 (b) and the remark following it.

The radical is $\text{nul}(\mathbf{A}) = \{\mathbf{y} : y_1 + 2y_2 + 3y_3 = 0\}$, with basis $(-2, 1, 0)$ and $(-3, 0, 1)$.

The left radical is $\text{nul}(\mathbf{A}^\top)$. Every row of $\mathbf{A}^\top$ is a multiple of $(1, 2, 1)$, so it is $\{\mathbf{x} : x_1 + 2x_2 + x_3 = 0\}$, with basis $(-2, 1, 0)$ and $(-1, 0, 1)$.

They are **not** equal: $(-3, 0, 1)$ lies in the radical but $-3 + 0 + 1 = -2 \neq 0$, so it is not in the left radical. Equal dimensions, different subspaces. In particular β is degenerate.

13.1.4.3 C. Going deeper

Exercise. C1: Over $\mathbb{F}_2$

Let $\beta(\mathbf{x}, \mathbf{y}) = x_1y_1 + x_2y_2$ on $V = \mathbb{F}_2^2$.

- Write down $[\beta]_{\mathcal{E}}^{\mathcal{E}}$ and prove that β is non-degenerate.
- Find every $\mathbf{v} \in V$ with $\beta(\mathbf{v}, \mathbf{v}) = 0$.
- Hence explain why the argument “ $\beta(\mathbf{v}, \mathbf{v}) = 0$ only for $\mathbf{v} = \mathbf{0}$ ” cannot be used to prove non-degeneracy over a general field, even for the dot product.

Solution.

- $[\beta]_{\mathcal{E}}^{\mathcal{E}} = \mathbf{I}_2$, which is invertible, so β is non-degenerate by Proposition 13.1.2.
- $\beta(\mathbf{v}, \mathbf{v}) = v_1^2 + v_2^2$. In $\mathbb{F}_2$ every element satisfies $t^2 = t$, so this is $v_1 + v_2$, which vanishes for $\mathbf{v} = (0, 0)$ and $\mathbf{v} = (1, 1)$. So the non-zero vector $(1, 1)$ satisfies $\beta(\mathbf{v}, \mathbf{v}) = 0$.
- Over $\mathbb{R}$, $\beta(\mathbf{v}, \mathbf{v}) = v_1^2 + v_2^2$ is a sum of squares and vanishes only at $\mathbf{0}$, so the implication holds there; the proof uses that a sum of non-negative reals is zero only when each term is. Over $\mathbb{F}_2$ there is no order, nothing is non-negative, and the conclusion is false for $(1, 1)$ even though the form is the dot product and is non-degenerate. Non-degeneracy is a statement about the rank of a matrix and survives the change of field; the diagonal statement is a statement about positivity and does not.

Exercise. C2: Orthogonal complements for a form

Let β be a bilinear form on a vector space V with $\dim V = n$, and for a subspace $U \subseteq V$ put

$$U^{\perp_\beta} := \{\mathbf{v} \in V : \beta(\mathbf{u}, \mathbf{v}) = 0 \text{ for every } \mathbf{u} \in U\}.$$

- Prove that $U^{\perp_\beta}$ is a subspace and that $\dim U^{\perp_\beta} \geq n - \dim U$.
- Prove that equality holds when β is non-degenerate.
- Give an example with $\dim U = 1$ in which the inequality in (a) is strict.

Hint: consider the map $V \rightarrow U^*$ that sends $\mathbf{v}$ to the restriction of $\beta(\cdot, \mathbf{v})$ to U .

Solution.

- (a) Let $\rho: V^* \rightarrow U^*$ be the restriction map and $S = \rho R_\beta: V \rightarrow U^*$, so $S(\mathbf{v}) = \beta(\cdot, \mathbf{v})|_U$. Both factors are linear, so S is linear, and $U^{\perp_\beta} = \ker S$ by the definition of S ; hence $U^{\perp_\beta}$ is a subspace. By Rank–Nullity (Theorem 3.3.1),

$$\dim U^{\perp_\beta} = n - \text{rank } S \geq n - \dim U^* = n - \dim U,$$

using $\text{im } S \subseteq U^*$ and $\dim U^* = \dim U$ (Corollary 4.1.4).

- (b) If β is non-degenerate then R_β is an isomorphism (Proposition 13.1.2 (c)), and ρ is surjective (Theorem 4.5.1). So $S = \rho R_\beta$ is surjective, $\text{rank } S = \dim U$, and the inequality in (a) is an equality.
- (c) Take $\beta(\mathbf{x}, \mathbf{y}) = x_1 y_1$ on F^2 and $U = \text{span}(\mathbf{e}_2)$. Then $\beta(\mathbf{e}_2, \mathbf{v}) = 0$ for every $\mathbf{v}$, so $U^{\perp_\beta} = F^2$ has dimension 2, while $n - \dim U = 1$. The form is degenerate, as (b) requires it to be.

Exercise. C3: The evaluation pairing

Let V be finite-dimensional and let $W = V^* \times V$. Define β on W by

$$\beta((\varphi, \mathbf{v}), (\psi, \mathbf{w})) := \psi(\mathbf{v}) + \varphi(\mathbf{w}).$$

- (a) Prove that β is a bilinear form on W .
- (b) Prove that β is non-degenerate. *Hint: the two summands of an element of the radical can be killed separately.*
- (c) Deduce that if $\mathbf{v} \in V$ satisfies $\varphi(\mathbf{v}) = 0$ for every $\varphi \in V^*$, then $\mathbf{v} = \mathbf{0}$, and name the result of Chapter 4 this recovers.

Solution.

- (a) Fix $(\psi, \mathbf{w})$. The map $(\varphi, \mathbf{v}) \mapsto \psi(\mathbf{v}) + \varphi(\mathbf{w})$ is the sum of $(\varphi, \mathbf{v}) \mapsto \psi(\mathbf{v})$, which is linear because ψ is and because projection onto the second summand is linear, and $(\varphi, \mathbf{v}) \mapsto \varphi(\mathbf{w}) = \text{ev}_{\mathbf{w}}(\varphi)$, which is linear because $\text{ev}_{\mathbf{w}} \in V^{**}$ (Theorem 4.3.1). So clause (B1) holds. Clause (B2) follows by the same argument with the roles of the two arguments exchanged, since the defining expression is symmetric in them.
- (b) Suppose $(\psi, \mathbf{w}) \in \text{rad}(\beta)$, so $\psi(\mathbf{v}) + \varphi(\mathbf{w}) = 0$ for all $\varphi \in V^*$ and all $\mathbf{v} \in V$. Taking $\varphi = 0$ gives $\psi(\mathbf{v}) = 0$ for every $\mathbf{v}$, so $\psi = 0$. Taking $\mathbf{v} = \mathbf{0}$ then gives $\varphi(\mathbf{w}) = 0$ for every $\varphi \in V^*$, so $\text{ev}_{\mathbf{w}} = 0$ in V^{**} , and $\mathbf{w} = \mathbf{0}$ because ev_V is injective (Theorem 4.3.1). Hence the radical is $\{\mathbf{0}\}$.
- (c) The second half of (b) is exactly that statement: $\varphi(\mathbf{v}) = 0$ for all $\varphi \in V^*$ forces $\mathbf{v} = \mathbf{0}$. It recovers the injectivity of the evaluation map $\text{ev}_V: V \rightarrow V^{**}$ of Theorem 4.3.1, which is the first half of Theorem 4.3.2. So the non-degeneracy of this form is the statement that functionals separate points (Theorem 4.1.5).